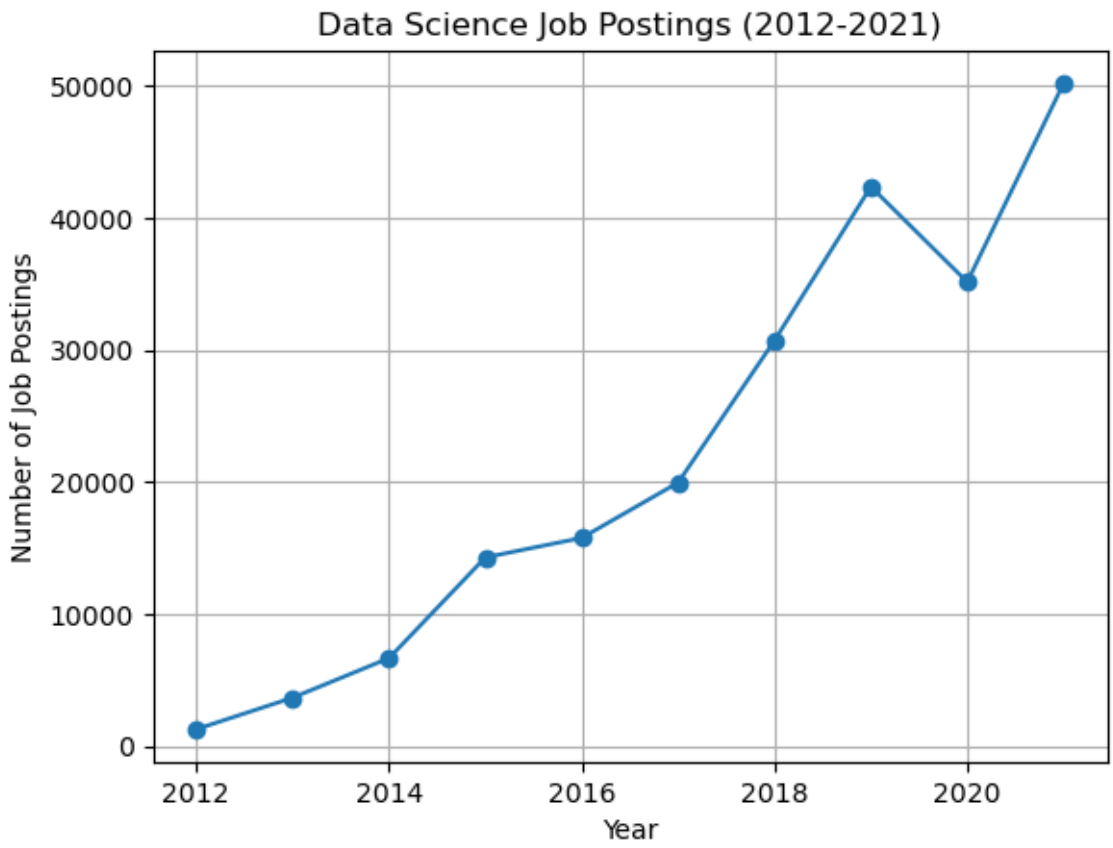
Exp No:1.a Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
 import pandas as pd
import matplotlib.pyplot as plt
year = [2012, 2013, 2014, 2015, 2016,
       2017, 2018, 2019, 2020, 2021]
jobs = [1260, 3649, 6644, 14272, 15749,
       19936, 30696, 42340, 35123, 50175]
df = pd.DataFrame({
   "Year": year,
   "Job Postings": jobs
})
print(df)
plt.plot(year, jobs, marker='o')
plt.xlabel("Year")
plt.ylabel("Number of Job Postings")
plt.title("Data Science Job Postings (2012-2021)")
plt.grid(True)
plt.show()
```

Output:



Exp No:1.b

Analyze and visualize the distribution of various data science roles (Data

Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize

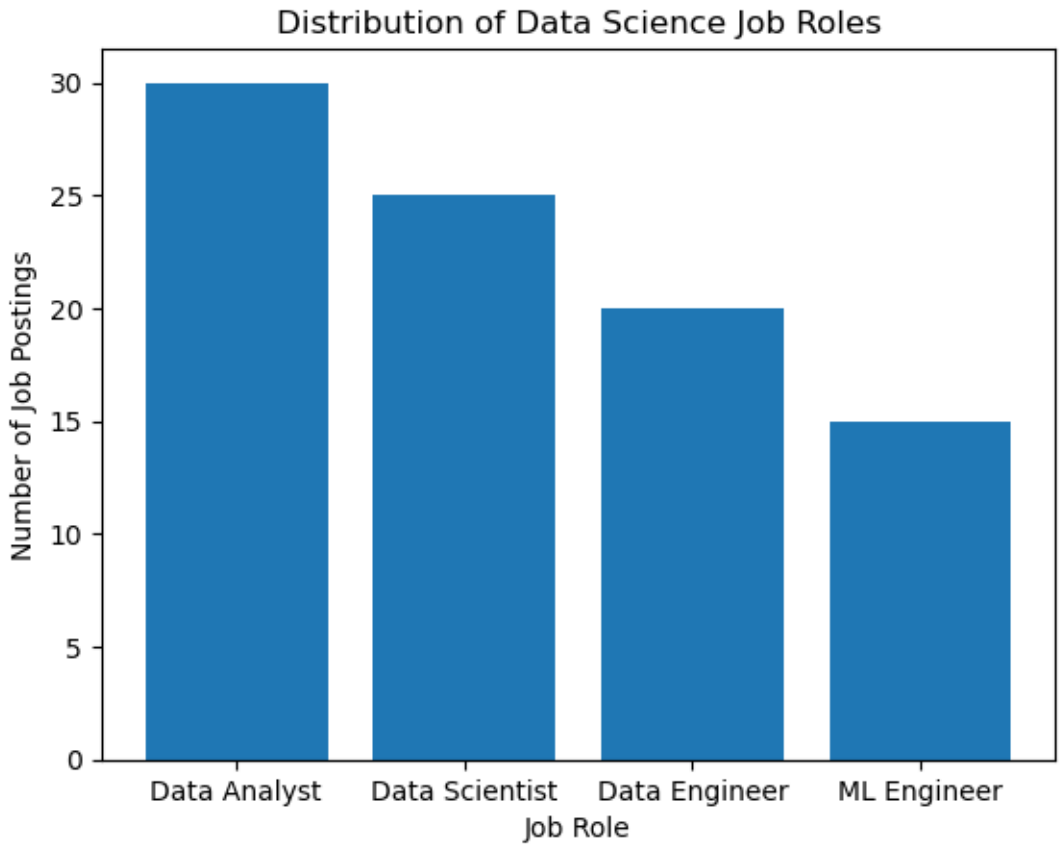
the distribution using pie charts or bar plots.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
roles = ["Data Analyst", "Data Scientist", "Data Engineer", "ML Engineer"]
count = [30, 25, 20, 15]
df = pd.DataFrame({
   "Job Role": roles,
   "Job Postings": count
})
```

```
print(df)
plt.bar(roles, count)
plt.xlabel("Job Role")
plt.ylabel("Number of Job Postings")
plt.title("Distribution of Data Science Job Roles")
plt.show()
```

Output:



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

Code:

```
import pandas as pd
structured = pd.DataFrame({
```

```
    "Name": ["Arun", "Bala", "Charan"],
    "Age": [20, 21, 22],
    "Marks": [85, 90, 78]
})
unstructured = [
    "Arun scored 85 marks.",
    "Bala scored 90 marks.",
    "Charan scored 78 marks."
]
semi_structured = [
    {"Name": "Arun", "Marks": 85},
    {"Name": "Bala", "Marks": 90},
    {"Name": "Charan", "Marks": 78}
]
print("Structured Data:")
print(structured)
print("\nUnstructured Data:")
print(unstructured)
print("\nSemi-Structured Data:")
print(semi_structured)
```

Output:

Structured Data:

	Name	Age	Marks
0	**Arun**	**20**	**85**
1	**Bala**	**21**	**90**
2	**Charan**	**22**	**78**

Unstructured Data:

['Arun scored 85 marks.', 'Bala scored 90 marks.', 'Charan scored 78 marks.']

Semi-Structured Data:

[{'Name': 'Arun', 'Marks': 85}, {'Name': 'Bala', 'Marks': 90}, {'Name': 'Charan', 'Marks': 78}]

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
cipher = Fernet(key)
data = "MyPassword123"
encrypted = cipher.encrypt(data.encode())
decrypted = cipher.decrypt(encrypted).decode()
print("Original Data:", data)
print("Encrypted Data:", encrypted)
print("Decrypted Data:", decrypted)
```

Output:

Original Data: MyPassword123

Encrypted Data: b'gAAAAABqgIq-jmngG39cEMkM50lFeHwhTwLvltrLJqGrUWi4HjF-nweYgBoEjojXTKni0dNLd-oYxUDRIW4jMSFO6qtijcioBQ=='

Decrypted Data: MyPassword123